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Unit 5 - Knowledge Check

PDF Documents

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KNOWLEDGE CHECK

1 What does IoT stand for?

- ☐ A Internet of Toys
- ☐ B Internet of Trees
- ☐ C Internet of Things
- ☐ D Internet of Televisions

CORRECT ANSWER

☒ C

2 Which of the following is an example of an IoT device?

- ☐ A A regular bicycle
- ☐ B A smart fridge that can connect to the internet
- ☐ C A wooden chair
- ☐ D A paper notebook

CORRECT ANSWER

☒ B

3 Why is it important to use eco-friendly solutions like solar panels in traffic lights?

- ☐ A They are cheaper than regular lights
- ☐ B They are more efficient in all weather conditions
- ☐ C They reduce carbon footprint and promote sustainability
- ☐ D They require less electricity from the grid

CORRECT ANSWER

☒ C

4 What is the purpose of incorporating an environmental module in the traffic light system?

- ☐ A To provide a sustainable power source and monitor energy levels
- ☐ B To simulate the effect of traffic on the environment
- ☐ C To enhance the aesthetic appeal of the traffic light
- ☐ D To collect data on environmental conditions

CORRECT ANSWER

☒ A

5 What is the main advantage of using a solar panel in the traffic light system?

- ☐ A It reduces the installation cost
- ☐ B It makes the traffic light look attractive
- ☐ C It provides a sustainable power source
- ☐ D It requires less maintenance

CORRECT ANSWER

☒ C

6 How does the energy level displayed change in the traffic light system?

- ☐ A It changes randomly
- ☐ B It changes based on user input
- ☐ C It changes based on the amount of light detected by the solar panel
- ☐ D It does not change

CORRECT ANSWER

☒ C

7 What does the modulus operator (%) do in programming?

- ☐ A Divides two numbers
- ☐ B Multiplies two numbers
- ☐ C Subtracts two numbers
- ☐ D Returns the remainder after division

CORRECT ANSWER

☒ D

Critical Thinking Challenge

- 1** If the solar panel detects different levels of sunlight based on the time of day, how can you program your traffic light system to adapt its operation accordingly? Create a plan outlining how you would use sensor data to modify the traffic light's timing throughout the day.

- 2** Suppose you want to check if a number is both even and a multiple of 5 using an if-else statement. How would you combine these conditions? Write out the logical expression you would use in your code and explain why it works.